

Estimating Infectious Disease Importation Risk during the 2026 FIFA World Cup

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The 2026 FIFA World Cup

Key facts

- **48 nations** — largest World Cup in history
- **3 co-host countries:** USA, Mexico, Canada
- **16 venues** across 11 US cities, 2 Canadian, 3 Mexican
- Matches from **June 11 – July 19, 2026**
- Expected ~**85 million** international arrivals to the US in June 2026 alone (NTTO forecast)
- Tickets sold to fans from all 48 qualified nations

Why does this matter?

Mass gathering events accelerate the **importation** of infectious diseases into host cities. Early quantification helps public health authorities prioritise **surveillance** and **preparedness** by venue.

Scientific question

*What is the probability that at least one case of dengue, malaria, measles, or pertussis is **imported** into each WC host city during June 2026 — and which source countries drive that risk?*

Four diseases studied

- **Dengue** — endemic in Latin America, SE Asia, Africa; strongly seasonal
- **Malaria** — sub-Saharan Africa, SE Asia, Middle East
- **Measles** — resurgent globally; under-vaccinated pockets
- **Pertussis** — globally endemic; early infectious phase easily missed

Three nested models

- ① **Baseline** — historical I-92 travel, regional incidence
- ② **WC-adjusted** — 2026 country-level travel surge
- ③ **Schedule-driven** — fan routing by match schedule

The Poisson importation framework

We model disease importation as a **Poisson process**.

If $\lambda_{h,d}$ is the *expected* number of imported cases of disease d arriving in city h , the probability of at least one importation is:

$$P(X_{h,d} \geq 1) = 1 - e^{-\Lambda_{h,d}} \quad (\text{Poisson})$$

where $\Lambda_{h,d} = \sum_s \lambda_{s,h,d}$ sums over all source units s .

Each source-unit contribution has the same structure across all three models:

$$\lambda_{s,h,d} = \underbrace{N_{s,h}}_{\text{arrivals}} \times \underbrace{I_{s,d}}_{\text{per-traveller infection prob.}} \times \underbrace{p_d}_{\text{travel while infectious}}$$

Disease-specific parameters

Disease	ρ_d	p_d	Rationale for p_d
Dengue	0.3	0.5	Mild/asymptomatic early phase
Malaria	0.2	0.3	Significant illness; partial immunity
Measles	0.6	0.05	High symptom burden; rare to travel
Pertussis	0.3	0.7	Catarrhal phase $\approx$ common cold

ρ_d — **reporting adjustment**: accounts for under-reporting and fraction of cases that are actively infectious.

p_d — **travel-while-infectious probability**: likelihood that an infected individual travels internationally during their infectious period.

Per-traveller infection probability: $I_{s,d} = \rho_d \cdot C_{s,d} / P_s$

Travel data

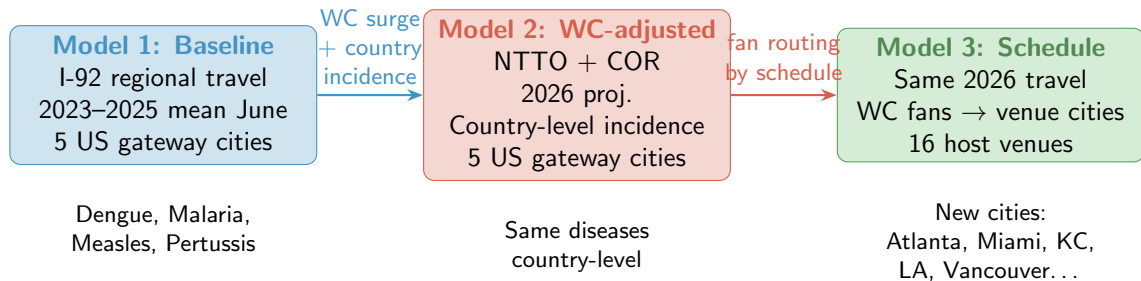
- **I-92 programme** (BTS/USDOC): monthly arrivals by world region to 6 US gateway cities (Boston, Dallas, Houston, Newark, New York, Philadelphia)
Sections 3, 8c, 9d
- **COR I-94 data** (CBP): monthly arrivals by country of residence — June 2024 used as baseline *Section 8b*
- **NTTO 2026 Forecast** (trade.gov): official 2026 projections for 12 top source markets *Section 8a*

Disease data

- **Dengue:** WHO Global Dengue Surveillance (monthly, accessed Dec 2025)
- **Malaria:** WHO Global Malaria Programme National Unit Data (2024 annual)
- **Measles:** WHO Immunization Data portal (annual, accessed Sep 2025)
- **Pertussis:** WHO Immunization Data portal (annual, accessed Dec 2025)

Match schedule: 69 games (group stage) across 16 venues from FIFA event records

Three-model hierarchy



M1 → **M2**: isolates the effect of the WC travel surge and country-level disease resolution.

M2 → **M3**: isolates the effect of schedule-based venue routing.

Model 1 — Baseline: equations

Travel volume: mean June arrivals over 2023–2025 from the I-92 programme, region r to city h : $N_{r,h}$

Routing fractions (reused in Models 2 & 3):

$$f_{r,h} = \frac{N_{r,h}}{N_{r, \text{US total}}} \quad (\text{routing})$$

Regional incidence:

$$I_{r,d} = \rho_d \cdot \frac{C_{r,d}}{P_r} \quad (\text{reg. incidence})$$

Importation intensity from region r to city h :

$$\lambda_{r,h,d} = N_{r,h} \cdot I_{r,d} \cdot p_d \quad \Lambda_{h,d} = \sum_r \lambda_{r,h,d} \quad (\text{baseline } \lambda)$$

Model 2 — WC-adjusted: travel volume construction

Tiered approach to 2026 travel volume:

Tier 1 — NTTO top-12 markets (Australia, Brazil, Canada, China, France, Germany, India, Italy, Japan, Mexico, South Korea, UK)

Growth factor $\phi_c = V_{c,2026}/V_{c,2024}$ from official NTTO Forecast Report. Already includes WC effect — *do not* add WC visitors on top.

Tier 2 — All remaining countries (COR I-94, June 2024)

- WC-qualified: $\phi_{WC} = 85,017/72,390 = \mathbf{1.174}$ (aggregate WC uplift)
- Non-qualified: $\phi_{base} = \mathbf{1.134}$ ($\approx 6.5\%/yr \times 2$ years)

$$N_{c, \text{Jun } 2026} = N_{c, \text{Jun } 2024}^{COR} \cdot \phi_c$$

$$N_{c,h}^{2026} = N_{c, \text{Jun } 2026} \cdot f_{r(c),h}$$

Model 2 — WC-adjusted: incidence and λ

Country-level incidence (more precise than regional):

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c} \quad (\text{country incidence})$$

Importation intensity:

$$\lambda_{c,h,d} = N_{c,h}^{2026} \cdot I_{c,d} \cdot p_d \quad \Lambda_{h,d} = \sum_c \lambda_{c,h,d} \quad (\text{WC-adj } \lambda)$$

What changes from Model 1:

- Travel volume reflects the WC surge and country-specific NTTO projections
- Incidence is country-level, not regional aggregate
- Cities remain the same 5 I-92 gateways (routing fractions are unchanged)

Model 3 — Schedule-driven: the core idea

Problem with Models 1 & 2

I-92 routing fractions assume WC fans distribute across cities the same way regular tourists do.

Reality: fans from Brazil go to *Houston* (where Brazil plays), not to Boston or Philadelphia.

Solution: separate the WC-specific travel increment from background tourism, and route fans using the match schedule.

Travel decomposition

$$N_c^{WC} = N_{c,24}^{COR} \cdot \max(0, \phi_c - \phi_{base})$$

$$N_c^{bg} = N_{c,Jun\ 2026} - N_c^{WC}$$

For WC-qualified countries:
WC increment $\approx 3.4\%$ of projected arrivals

$$\frac{1.174 - 1.134}{1.174} \approx 0.034$$

Model 3 — Schedule-driven: routing equations

Schedule-based WC fan routing fraction:

$$\omega_{c,v} = \frac{g_{c,v}}{G_c} \quad (\text{schedule routing})$$

$g_{c,v}$ = games played by country c 's team at venue city v ; G_c = total known group-stage games.

Arrival matrices:

$$N_{c,v}^{\text{WC}} = N_c^{\text{WC}} \cdot \omega_{c,v} \quad (\text{WC fans} \rightarrow \text{venue cities})$$

$$N_{c,h}^{\text{bg}} = N_c^{\text{bg}} \cdot f_{r(c),h} \quad (\text{background} \rightarrow \text{I-92 cities})$$

Combined importation intensity for venue city v :

$$\Lambda_{v,d} = \underbrace{\sum_{c \in Q} N_{c,v}^{\text{WC}} \cdot I_{c,d} \cdot p_d}_{\text{WC fans}} + \underbrace{\sum_c N_{c,h(v)}^{\text{bg}} \cdot I_{c,d} \cdot p_d}_{\text{background}} \quad (\text{schedule } \lambda)$$

Model 3 — Venue coverage

City	Country	I-92?
New York [†]	USA	Yes
Dallas [†]	USA	Yes
Houston [†]	USA	Yes
Philadelphia [†]	USA	Yes
Boston [†]	USA	Yes
Los Angeles	USA	No
Atlanta	USA	No
Kansas City	USA	No
Miami	USA	No
San Francisco	USA	No
Seattle	USA	No
Vancouver	Canada	No
Toronto	Canada	No
Mexico City	Mexico	No
Monterrey	Mexico	No
Guadalajara	Mexico	No

Key insight

5 I-92 cities: both WC-fan and background streams contribute.

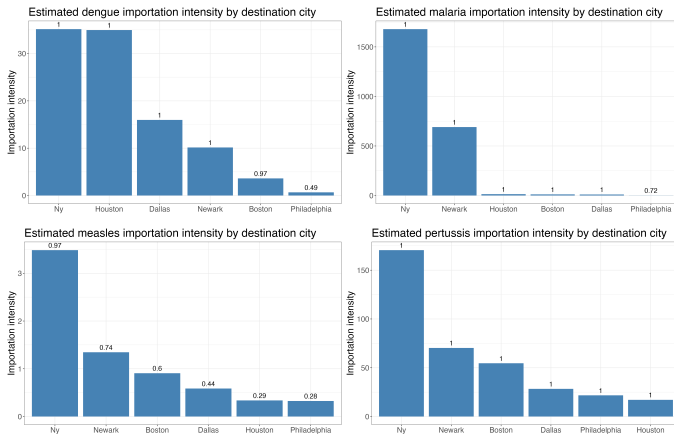
11 other venues: only the WC-fan stream contributes.

These cities are invisible in Models 1 & 2.

England + Scotland issue:

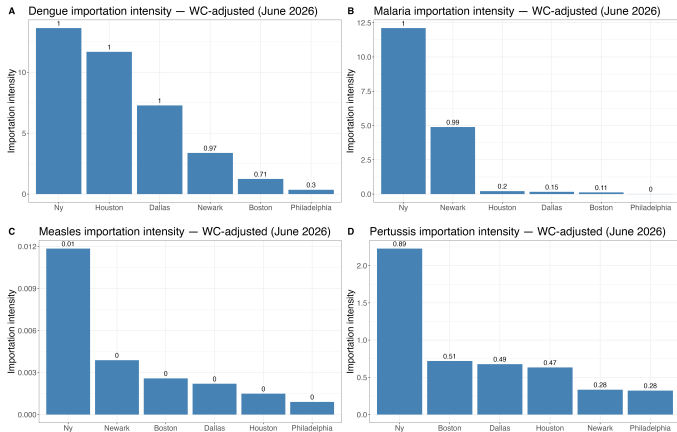
Both map to “United Kingdom” in COR data. Their games are combined; routing fractions sum to 1 — no double-counting.

Model 1 — Baseline results



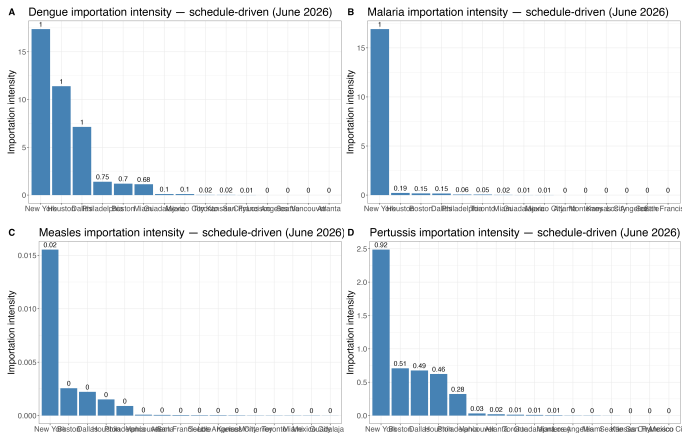
Numbers above bars = $P(\geq 1 \text{ importation})$; bar height = expected importation intensity λ .

Model 2 — WC-adjusted results



Same 5 US gateway cities; travel volumes now reflect June 2026 projections and country-level disease incidence.

Model 3 — Schedule-driven results



Up to 16 venue cities visible; WC fans are routed to where their team plays; background travelers use I-92 routing.

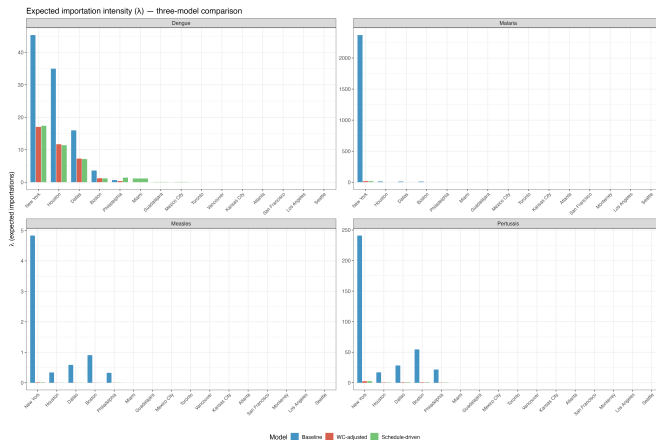
Three-model comparison — importation probability



■ Baseline ■ WC-adjusted ■ Schedule-driven.

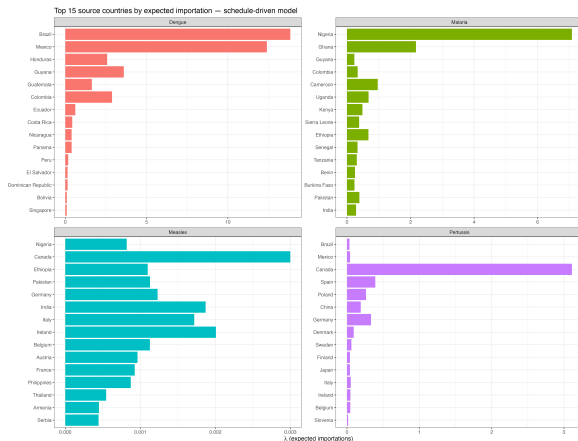
$P(\geq 1)$ saturates to 1 for high-risk cities — see next slide for the linear-scale λ comparison.

Three-model comparison — importation intensity (λ)



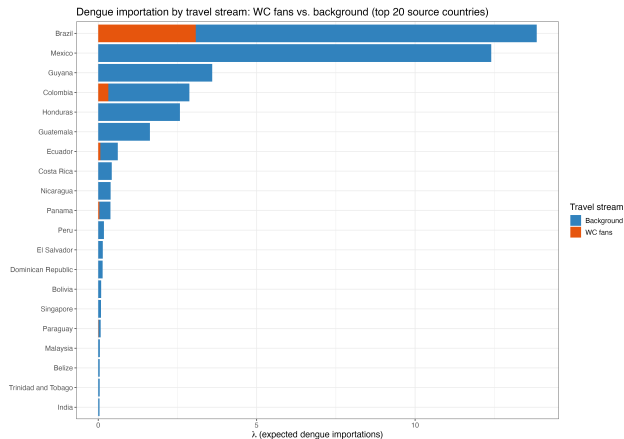
λ stays linear even when $P(\geq 1) \rightarrow 1$, revealing relative risk more clearly than the probability scale.

Which countries drive importation risk?



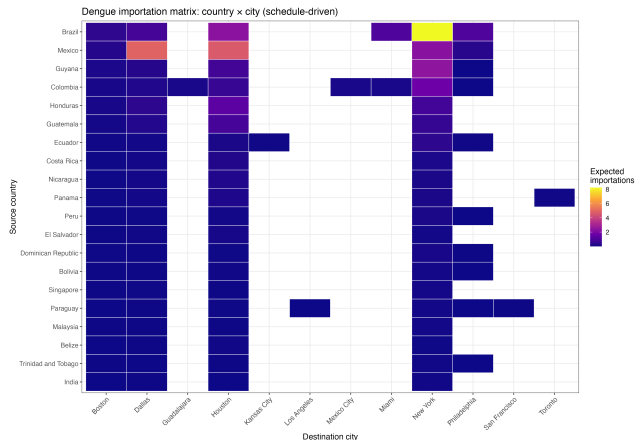
Top 15 source countries by total expected importations (schedule-driven model, summed across all destination cities).

WC fans vs. background tourists — dengue



■ WC fans ■ Background. Countries with WC teams show a non-zero WC-fan component; non-qualified countries contribute only through background travel.

Country × city importation matrix — dengue



Colour = expected dengue importations $\lambda_{c,v}$. WC-qualified countries show concentration in specific match venues; non-qualified countries show diffuse contributions across I-92 cities.

Key findings

- 1 **Dengue dominates** importation intensity across all three models and all destination cities, driven by high incidence in Latin America and SE Asia.
- 2 **The WC travel surge** (Model 1 $\rightarrow$ 2) increases estimated importation intensity for qualified nations but city rankings remain broadly stable.
- 3 **Schedule-driven routing** (Model 2 $\rightarrow$ 3) reveals risk at *new* venue cities (Atlanta, Miami, Kansas City, Los Angeles, Mexican venues) that are invisible to I-92-based models.
- 4 **Background travel** dominates the total importation burden for most diseases; the WC-fan increment is $\approx 3.4\%$ of projected arrivals for qualified countries.
- 5 **A small number of high-incidence countries** (Brazil, Colombia, and others by disease) account for a disproportionate share of city-level risk.

Limitations

Parameter uncertainty

- ρ_d and p_d are central estimates — no strong empirical constraint for mass sporting events
- Sensitivity analysis ($\pm 50\%$ on both parameters) recommended before policy communication

Travel data coverage

- I-92 routing fractions available only for 5 US gateway cities
- 11 other WC venues receive only the WC-fan stream — background risk is underestimated there

Model scope

- Importation only — no secondary transmission modelled
- Knockout-round games excluded (TBD opponents)
- Fan routing assumes proportional allocation by games; actual travel behaviour may differ

Data recency

- Disease data through late 2025; current outbreak situations may differ
- NTTD projections pre-date the event; actual arrival volumes will vary

Data that would most improve the model

Travel volume

- **Flight bookings** (OAG, Amadeus, IATA PaxIS) — direct country-to-city routing for the WC window; covers the 11 venues not in I-92
- **FIFA ticket sales** by country — most direct measure of WC fan travel; replaces the NTTO decomposition entirely
- **Previous WC entry records** (Brazil 2014, Russia 2018, Qatar 2022) — empirical validation of schedule-based fan routing

Disease burden

- **Real-time surveillance** (ProMED, WHO DON, ECDC CDTR) — catch outbreaks that emerge after the 2025 data freeze
- **Age-stratified incidence** — travelers skew working-age; age-specific burden would refine $I_{c,d}$
- **Vaccination coverage** (WHO/UNICEF) — adjust effective measles/pertussis incidence for immunity in travelers

Local transmission

- **Ae. aegypti surveillance** (CDC ArboNET) — extend model from importation to local dengue outbreak probability
- **Hospital capacity** (AHA survey) — contextualise risk for health system resource pre-positioning
- **Mobility data** (Meta Movement, Google) — validate the schedule-based fan routing assumption empirically

Next steps

- ➊ **Sensitivity analysis** — systematic grid over (ρ_d, p_d) space; produce uncertainty intervals around λ and $P(\geq 1)$
- ➋ **Airport-level allocation** — distribute city-level importation risk across individual US airports using BTS enplanement data and spatial join to counties (code already in supplementary file)
- ➌ **Incorporate knockout rounds** — as the WC bracket becomes known, update the schedule-driven model with resolved match assignments
- ➍ **Real-time surveillance linkage** — compare model predictions against actual case notifications from CDC and partner health agencies during the tournament
- ➎ **Expand disease scope** — cholera, mpox, and influenza are under consideration for a follow-up analysis

Thank you

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Code and data:

`https://github.com/jdiestra1977/FIFA\_worldCup\_2026\_risk`